

CIS 371 Web Application Programming

Introduction



Lecturer: **Dr. Haoyu Li**



What is Web Application Programming?

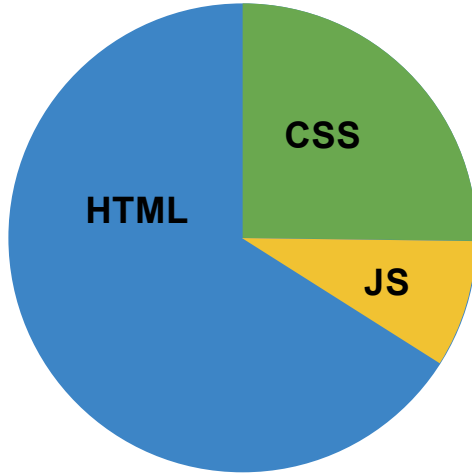
What is Web Application Programming?

Web Apps $\neq$ Web Pages



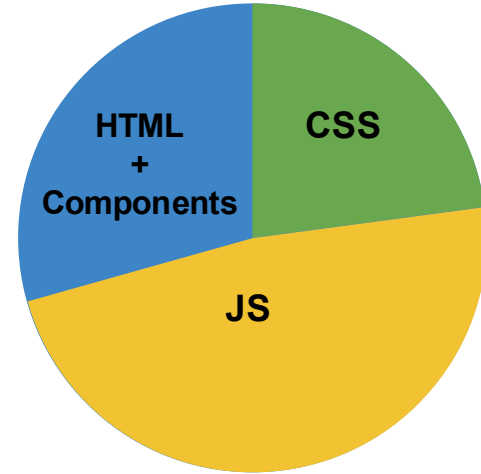
CS 371 (this course)

Web Pages vs. Web Apps



Static Web Pages

Web 1.0: read-only web



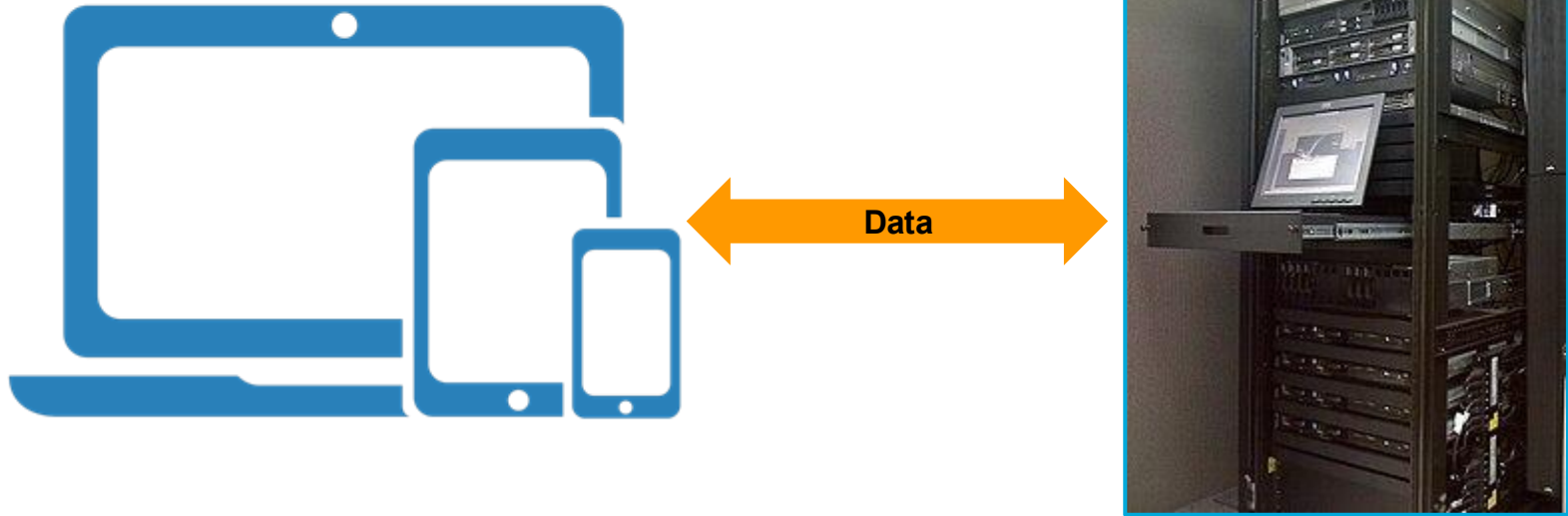
Web Apps

Web 2.0: dynamic read-write web

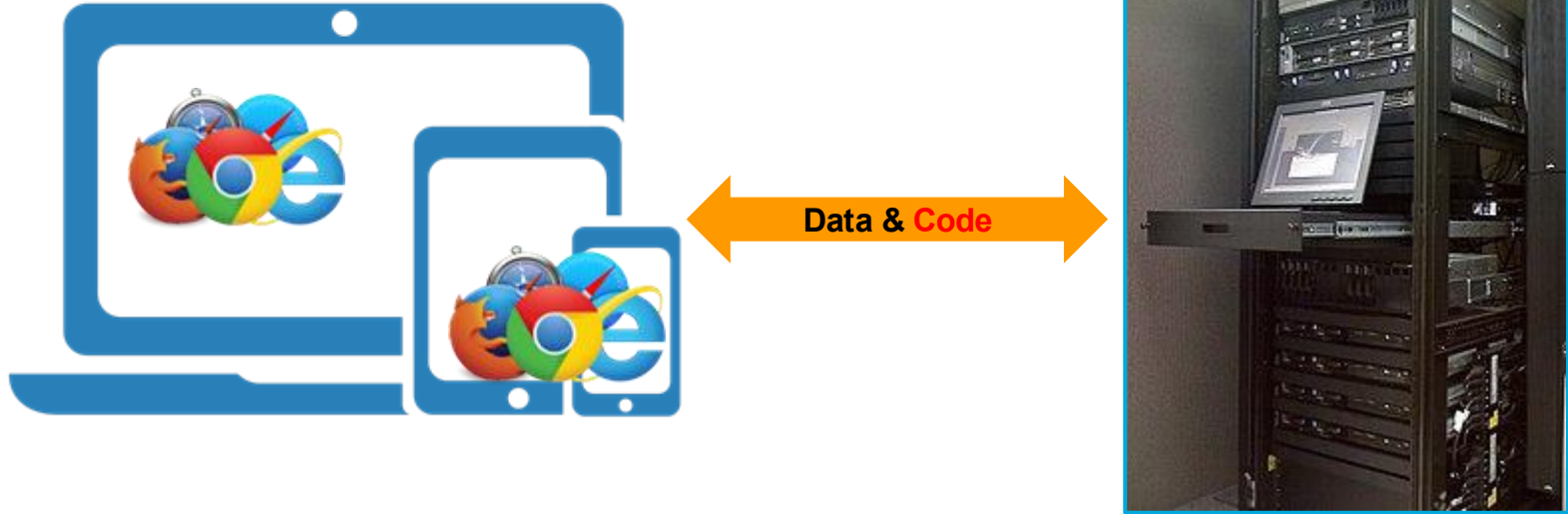
Desktop / Mobile Apps vs. Web Apps

Desktop / Mobile Apps

Traditional Client/Server architecture



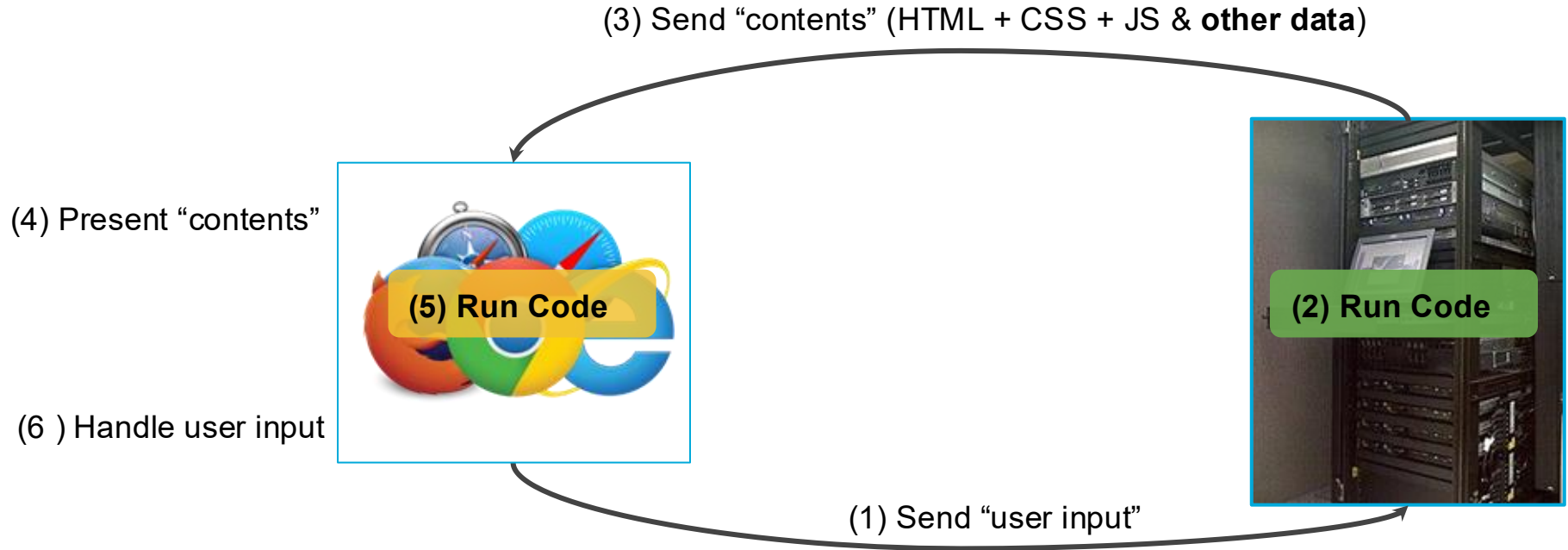
Web Apps: Client/Server architecture



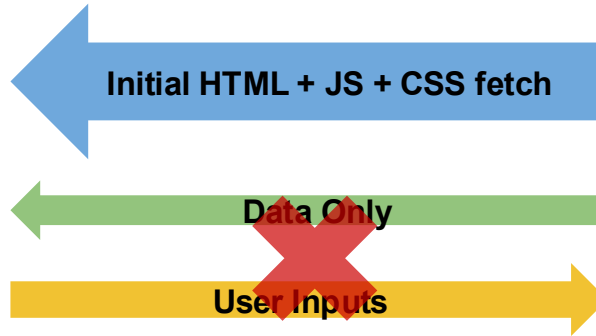
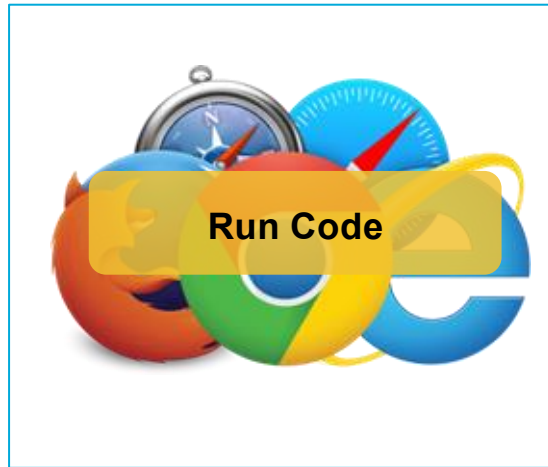
Roles of Web Browsers in Web apps

- Present data
 - HTML + Text + Audio + Video + Image
 - Content animation (CSS)
 - 2D Graphics or 3D Graphics (WebGL) on <canvas>
- Receiver user input
 - Textual input
 - Mouse clicks / screen taps
 - Screen orientation (gyroscope on smartphones) ⇒ WebVR
- Run code
 - JavaScript (engines: Google V8, Mozilla SpiderMonkey, Apple JSCore, Microsoft Chakra)
 - Web Assembly (proposal since 2017)

Web Client/Server Architecture



Web 1.0: Static web pages



Example of Web 1.0

- Home of the first website: <https://info.cern.ch/>

World Wide Web

The WorldWideWeb (W3) is a wide-area [hypermedia](#) information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an [executive summary](#) of the project, [Mailing lists](#), [Policy](#), November's [W3 news](#), [Frequently Asked Questions](#),

[What's out there?](#)

Pointers to the world's online information, [subjects](#), [W3 servers](#), etc.

[Help](#)

on the browser you are using

[Software Products](#)

A list of W3 project components and their current state. (e.g. [Line Mode](#), [X11](#), [Viola](#), [NeXTStep](#), [Servers](#), [Tools](#), [Mail robot](#), [Library](#))

[Technical](#)

Details of protocols, formats, program internals etc

[Bibliography](#)

Paper documentation on W3 and references.

[People](#)

A list of some people involved in the project.

[History](#)

A summary of the history of the project.

[How can I help?](#)

If you would like to support the web..

[Getting code](#)

Getting the code by [anonymous FTP](#), etc.



Web 2.0: Client/Server (dynamic R/W web apps)

